

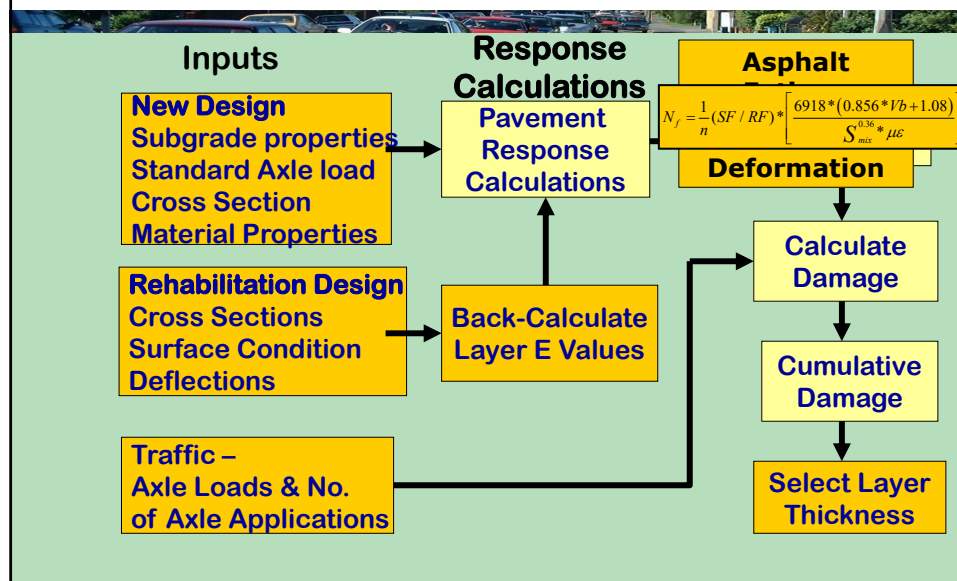
Mechanistic-Empirical Flexible Pavement Design Procedures

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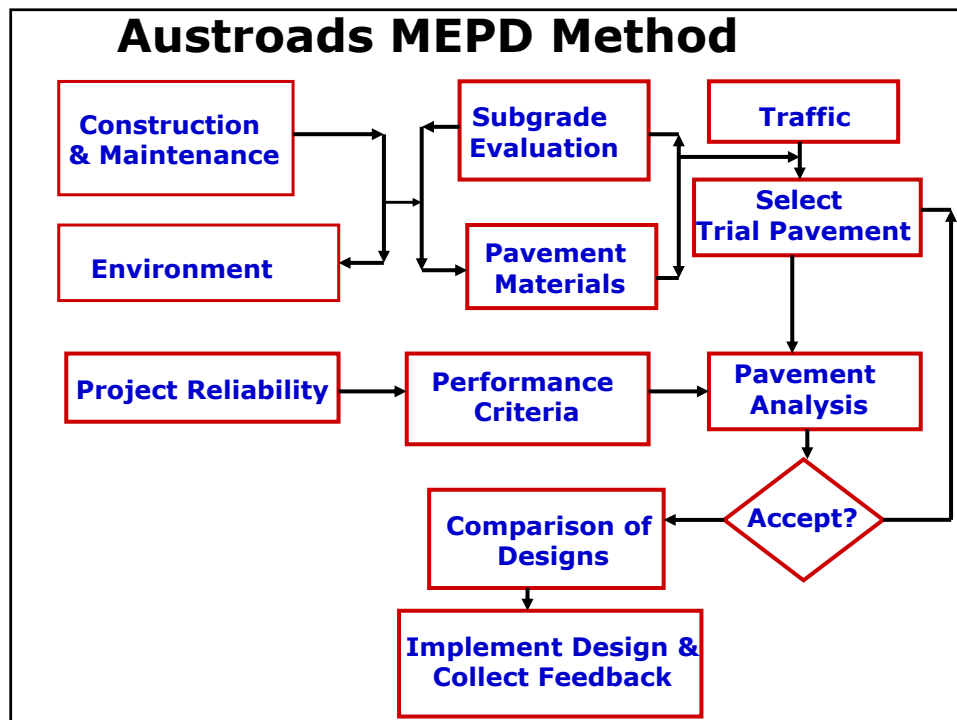
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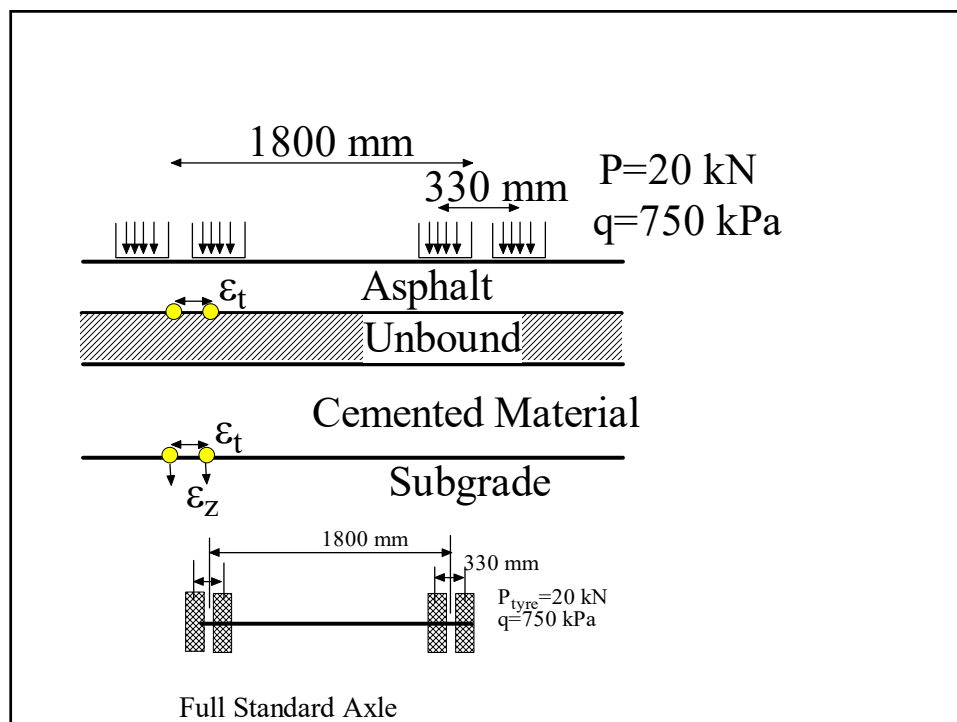
Austrroads ME Pavement Design



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Failure Criteria

$$N_f = \frac{1}{n} \frac{SF}{RF} * \left[\frac{6918 * (0.856 * Vb + 1.08)}{E_{asphalt}^{0.36} * \mu\epsilon} \right]^5$$

Asphalt Concrete

$$N_f = \left[\frac{9150}{\mu\epsilon} \right]^7$$

**Subgrade
criteria**

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Cemented materials

$$N_{ij} = \frac{1}{n} * RF * \left(\frac{K}{\mu\epsilon_{ij}} \right)^{12}$$

N_{ij}= allowable Number of Load repetitions
μ_{ε_{ij}}= Tensile strain for axle i applying load j
K=SF*k=1.55*k
k= laboratory fatigue constant
RF= Reliability factor

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Suggested reliability factors (RF) for cemented materials fatigue

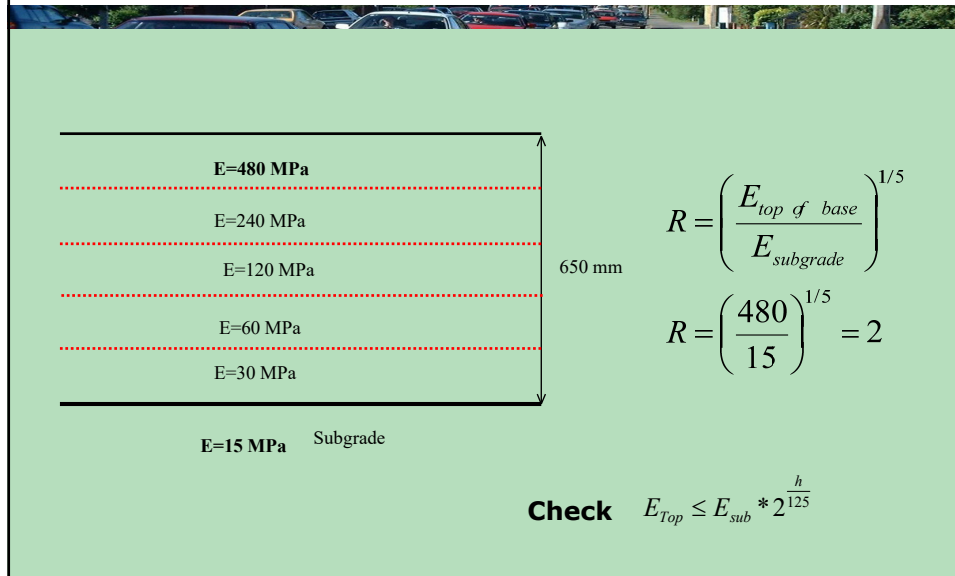
Project Reliability					
50	80	85	90	95	97.5
25	4.7	3.3	2	1	0.5

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Stress dependency and Sub layering approach

- Moduli of granular and selected subgrade materials are dependent not only on the intrinsic properties of these materials, but also on the stress level at which they operate and the stiffness of the underlying materials.
- For granular materials placed directly on cemented subbase, no sublayering is required.
- For granular materials placed directly on the insitu subgrade or selected subgrade materials, sublayering should be conducted.

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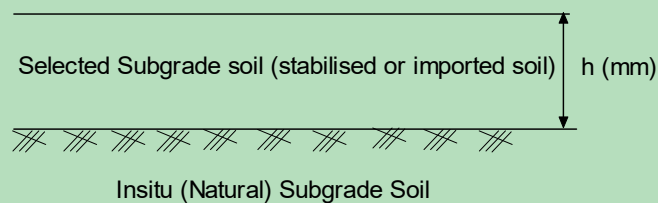
Selecting Subgrade Modulus

- The vertical modulus of the top selected subgrade layer is the minimum of these two values:

$$M_r = 10 * CBR_{(selected \text{ subgrade})}$$

$$M_r(\text{Selected subgrade top layer}) = M_{r(\text{insitusubgrade})} * 2^{(h/150)}$$

$$h = \text{Total selected subgrade thickness (mm)}$$



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Selecting Basecourse Modulus

- The vertical modulus of the top granular base layer is the smallest of these three values:
- Measured Resilient modulus in the lab or the presumptive value given in Table, or

$$M_{r(\text{Top Granular sublayer})} = M_{r\text{subgrade}} * 2^{(h/125)}$$

$h = \text{granular layer thickness (mm)}$

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Table 6.4(a) Suggested vertical modulus (MPa) of top sublayer of normal standard base material

Thickness of overlying material	Modulus of cover material* (MPa)				
	1000	2000	3000	4000	5000
40 mm	350	350	350	350	350
75 mm	350	350	340	320	310
100 mm	350	310	290	270	250
125 mm	320	270	240	220	200
150 mm	280	230	190	160	150
175 mm	250	190	150	150	150
200 mm	220	150	150	150	150
225 mm	180	150	150	150	150
≥250 mm	150	150	150	150	150

* Cover material is either asphalt or cemented material or a combination of these materials.

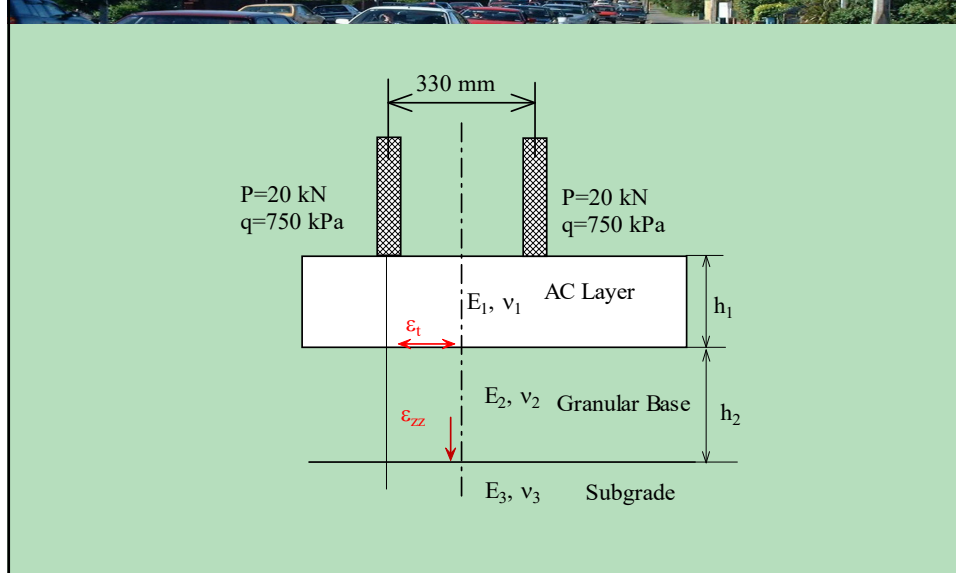
Table 6.4(b) Suggested vertical modulus (MPa) of top sublayer of high standard base material

Thickness of overlying material	Modulus of cover material* (MPa)				
	1000	2000	3000	4000	5000
40 mm	500	500	500	500	500
75 mm	500	500	480	460	440
100 mm	500	450	410	390	360
125 mm	450	390	350	310	280
150 mm	400	330	280	240	210
175 mm	360	270	210	210	210
200 mm	310	270	210	210	210
225 mm	260	210	210	210	210
≥250 mm	210	210	210	210	210

* Cover material is either asphalt or cemented material or a combination of these materials.

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AUSTROADS Design Method



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Cumulative Damage Factor

$$CDF = \sum_1^i \sum_1^j \frac{n_{ij}}{N_{ij}}$$

n_{ij} = Cumulative design traffic for axle load group i and axle load j

N_{ij} = Allowable cumulative traffic for axle group i and axle load j

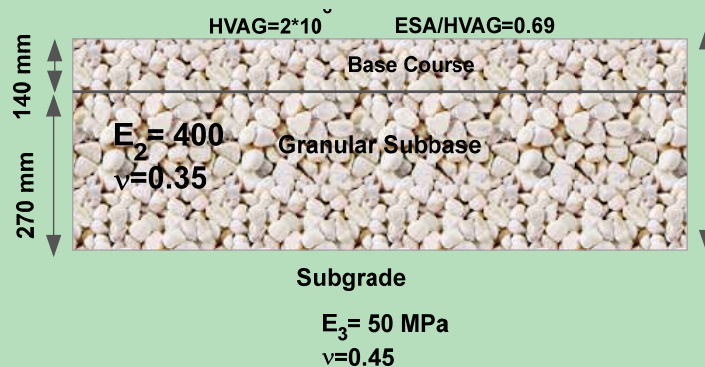
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AUSTROAD Design Method

- Used the design charts, and Circly to determine the thickness of flexible pavement to carry 2×10^6 HVAG, and the $ESA/HVAG=0.69$, if the subgrade CBR is 5% for the following cases:
 1. Chip seal over unbound base
 2. 150 mm AC layer over unbound material
 3. 150 mm of AC layer over 170 mm granular materials over cement treated base
 4. Full depth asphalt

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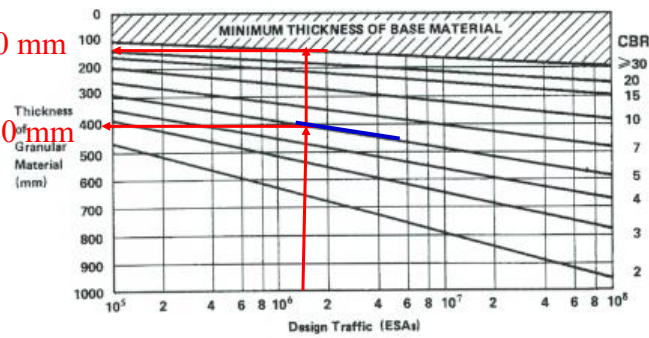
$$DES_{A} = N_{HVAG}^* \\ ESA/HVAG = 2 \times 10^6 \times 0.69 = 1.38 \times 10^6$$



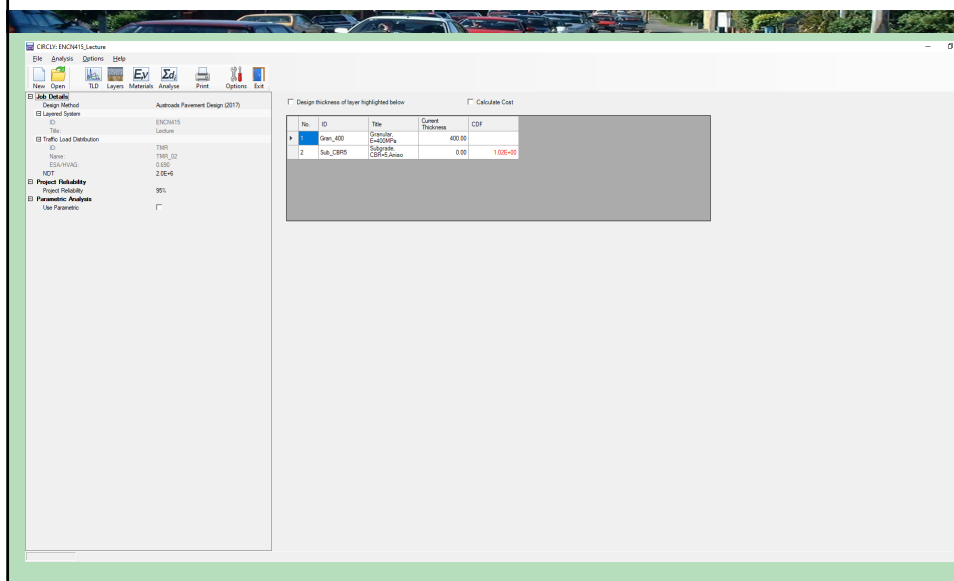
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140 mm

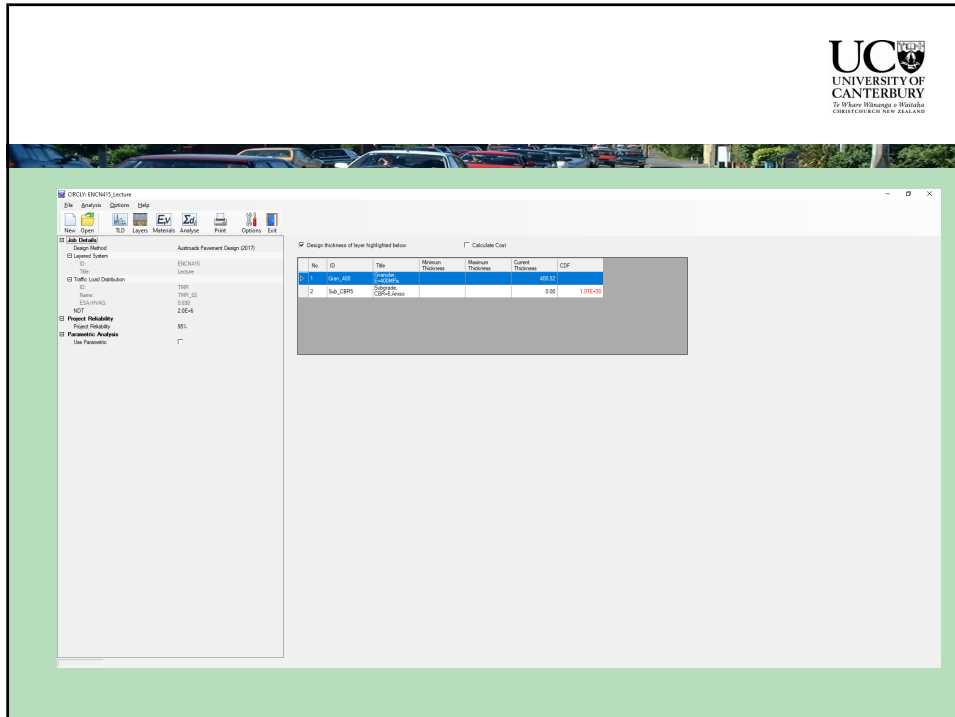
410 mm



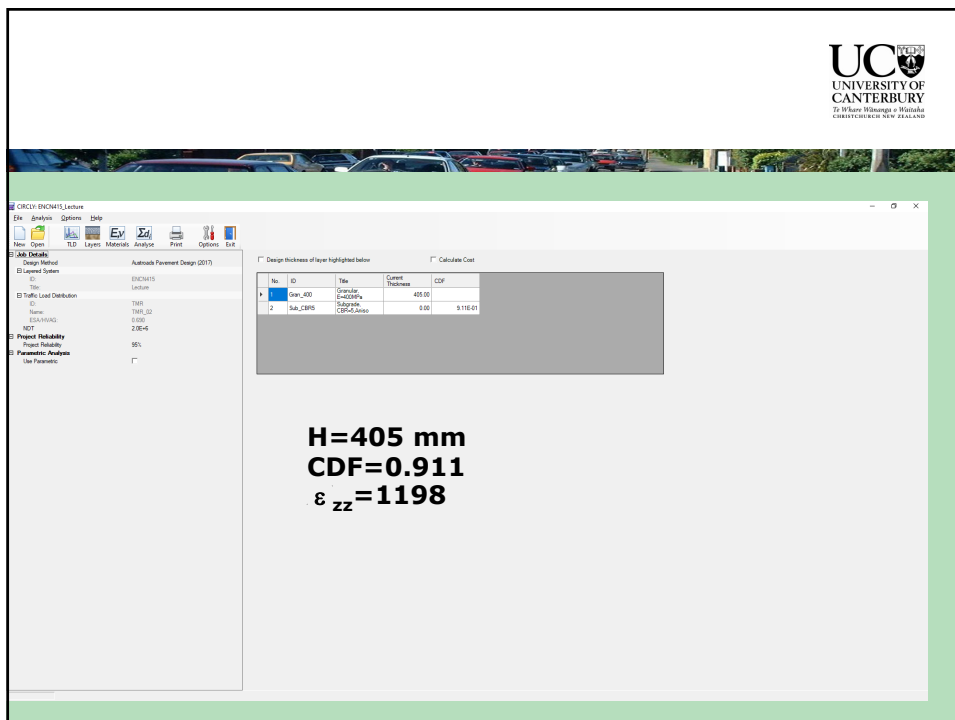
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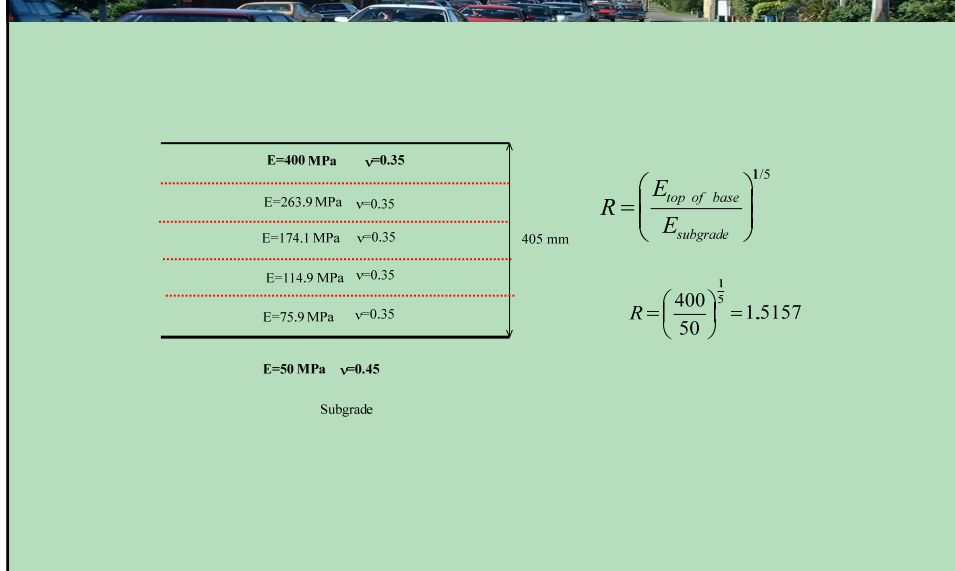


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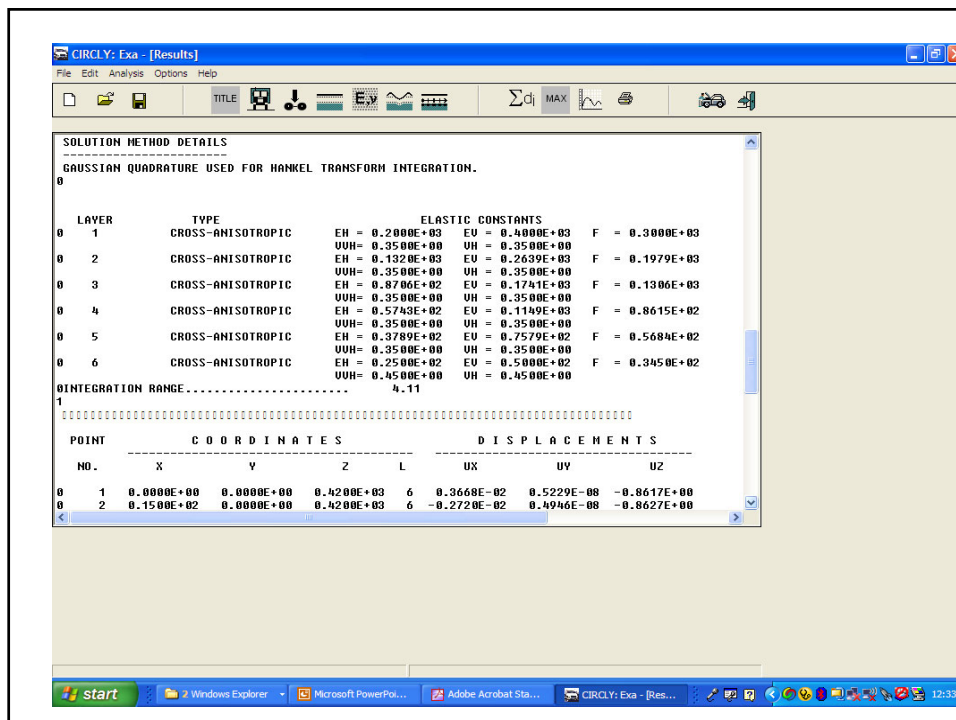


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Check Using Mechanistic Analysis



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- $\varepsilon_{zz} = 1198 \mu\epsilon$

$$N_f = \left(\frac{9150}{1198} \right)^7 = 1.52 * 10^6$$

$$CDF = \frac{0.69 * 2 * 10^6}{1.52 * 10^6} = 0.908$$

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Heavier traffic Spectrum

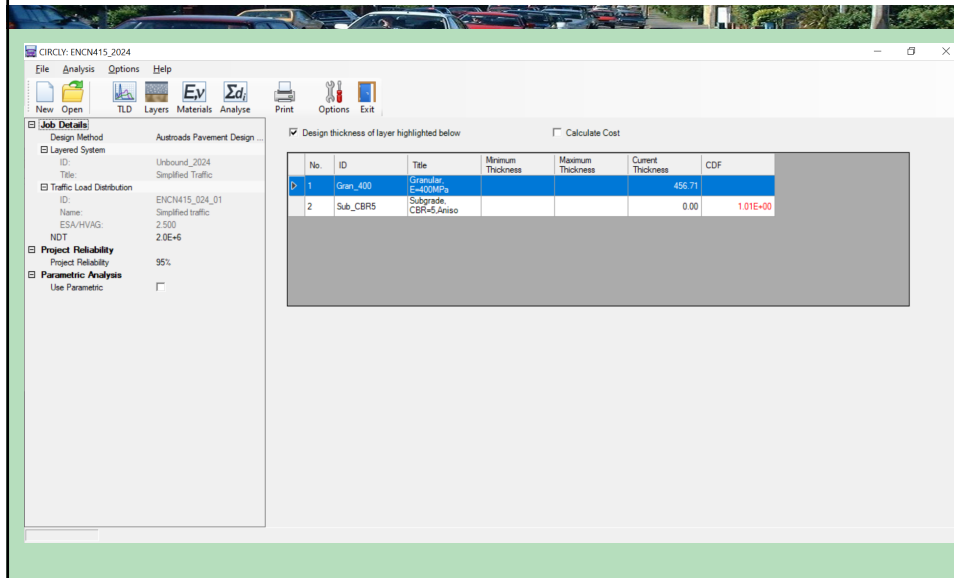
- Traffic Load Distribution

Load kN	SAST	SADT	QADT
85	0.3	0.4	0.3

$$ESA / HVAG = 0.3 * \left(\frac{85}{53} \right)^4 + 0.4 * \left(\frac{85}{80} \right)^4 + 0.3 * \left(\frac{85}{221} \right)^4 = 2.501$$

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Designing for Heavier Axle loads



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Asphalt Fatigue Calculations

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Te Whare Wānanga o Hātinaka
CHRISTCHURCH NEW ZEALAND

- Traffic Load Distribution

Load kN	SAST	SADT	QADT
85	0.3	0.4	0.3

$$ESA / HVAG = 0.3 * \left(\frac{85}{53}\right)^4 + 0.4 * \left(\frac{85}{80}\right)^4 + 0.3 * \left(\frac{85}{221}\right)^4 = 2.501$$

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Fatigue Parameters

- Asphalt Fatigue
- Fatigue Constant calculation:

$$N_f = \frac{1}{n} * \frac{SF}{RF} \left[\frac{6918 * (0.856 * Vb + 1.08)}{E_{asphalt}^{0.36} * \mu \epsilon_{ij}} \right]^5 = \frac{1}{n} * \frac{SF}{RF} \left(\frac{C}{\mu \epsilon} \right)^5$$

$$C = 6918 * (0.856 * 11 + 1.08) / 3000^{0.36} = 4066.7$$

Make sure that the performance parameters in Circly is adjusted so that the constant is (C=0.004067) and Exponent is 5.0

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Reliability Factors

Desired project reliability					
50%	80%	85%	90%	95%	97.5%
1.0	2.4	3.0	3.9	6.0	9.0

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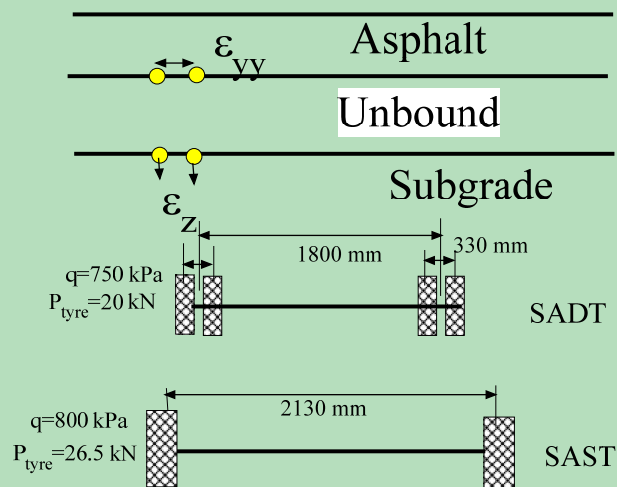
Design Traffic in HVAG

- $NDT = 2 \times 10^6$
- Number of application of each Axle Group

Load kN	SAST	SADT	QADT
85	0.3	0.4	0.3
nij	6×10^5	8×10^5	6×10^5

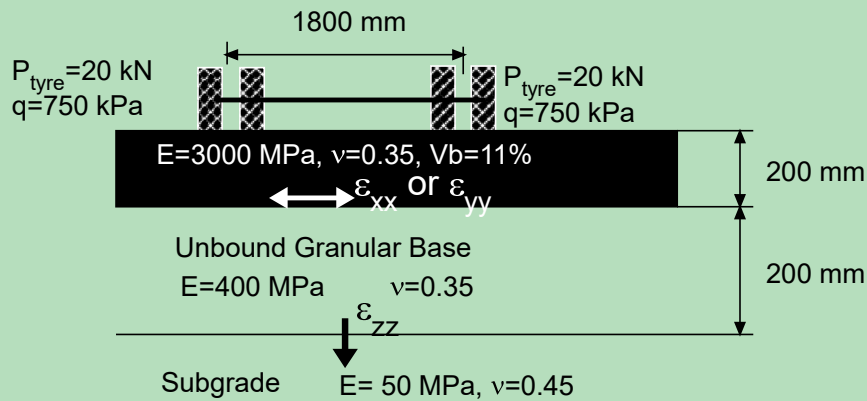
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APPLY Standard SADT and SAST



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Using Circly to Model the Pavement



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Circly Output

Asphalt thickness=200 mm
CDF (Asphalt)=0.857
CDF(Subgrade)= 4.89×10^{-3}

▪ Critical Responses under Standard Axles

Axle	SAST	SADT
Bottom of AC	160.2	192
Top of Subgrade	363.2	472.4

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Damage Calculations in Circly

- CDF for subgrade=

$$N_f = \left(\frac{9150}{472.4} \right)^7 = 1.023 * 10^9$$

$$CDF = \frac{2.5 * 2 * 10^6}{1.023 * 10^9} = 4.889 * 10^{-3}$$

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Damage Calculation For Asphalt Layer

- Scaling strains for all axles in the traffic Load Distribution

$$\mu\epsilon_{ij} = \frac{L_{ij}}{n} * \frac{\mu\epsilon_{SAST,53}}{53}$$

$$\mu\epsilon_{ij} = \frac{L_{ij}}{n} * \frac{\mu\epsilon_{SADT,80}}{80}$$

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Critical Strains

Axle	SAST	SADT
Bottom of AC	160.2	192
Top of Subgrade	363.2	472.4

AXLe Group i	SAST	SADT	QADT
Load (j)	SAST	SADT	QADT
85	256.93	204	51

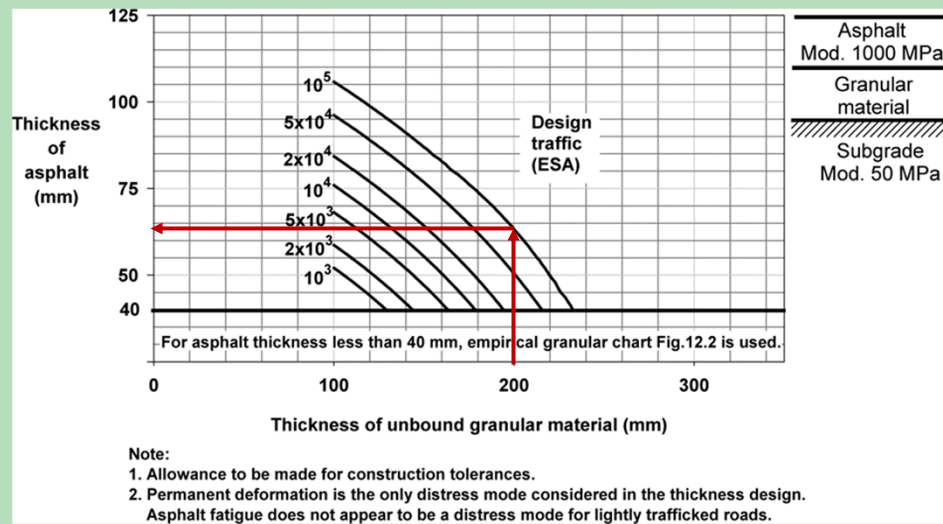
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$$N_{ij} = \frac{1}{n} * \frac{SF}{RF} * \left(\frac{0.0040667}{\epsilon_{ij}} \right)^5$$

Load kN	SAST	SADT	QADT	Total
ϵ_{ij}	$256.93 * 10^{-6}$	$204 * 10^{-6}$	$51 * 10^{-6}$	
n_{ij}	$6 * 10^5$	$8 * 10^5$	$6 * 10^5$	$2 * 10^6$
N_{ij}	$9.934 * 10^5$	$3.1482 * 10^5$	$8.0593 * 10^8$	
CDF	0.60397	0.25411	$7.4448 * 10^{-4}$	0.8588

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Design by Chart



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Cement Treated Base

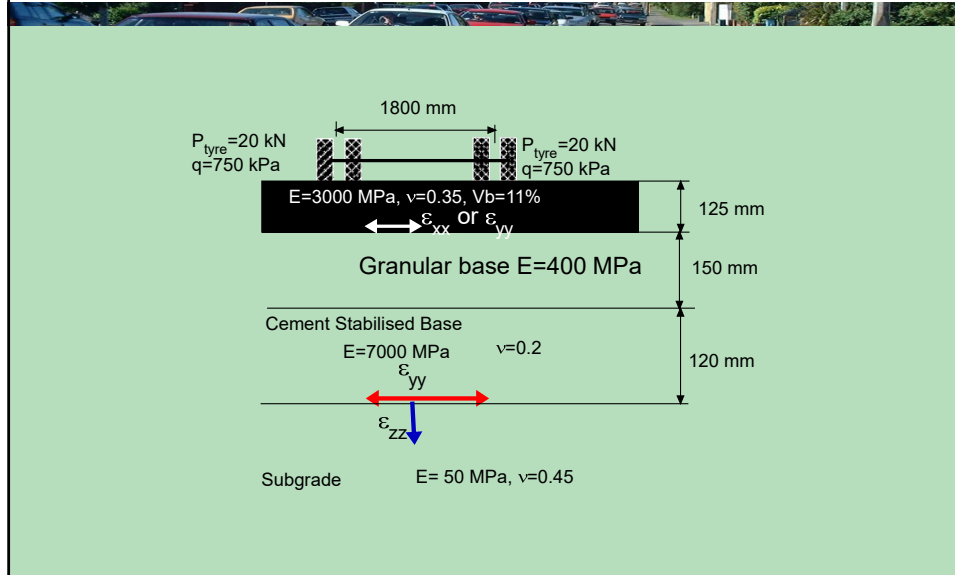
$$N_{ij} = RF * \left(\frac{K}{\mu \epsilon_{ij}} \right)^{12}$$

Table 6.19: Presumptive fatigue constants for lean-mix concrete

Property	Lean rolled concrete	Lean screeded concrete
Typical modulus (MPa)	7000	10 000
In-service fatigue constant K	242	223

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Cement Stabilised Alternative



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Damage calculations for CTB

Damage calculation for CTB is in the same way as asphalt.

$$CDF = \sum_1^j \sum_1^i \frac{n_{ij}}{N_{ij}}$$

n_{ij} = Cumulative design traffic for axle load group j and axle load i

N_{ij} = Allowable cumulative traffic for axle group j and axle load i

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Circly Output

Asphalt damage CDF=0.803
CTB Damage CDF= 1.03
Subgrade Damage CDF= 4.15×10^{-5}

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Circly Design

- Failure Criteria:
- Asphalt Fatigue
- Cement Stabilised fatigue
- Subgrade Rutting
- The constant of the asphalt fatigue is as before.

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Summary

- M-E Design Methodology provides a method for tying mixture and structural design together.
- M-E design methodology provides a method for considering multiple distresses.

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Questions?



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